

SIMULOS® Complete Standards & Modules Index

Simulation Governance Architecture — Simulation Governance Layer

This document serves as the official OOF® simulation governance architecture map and structural index used for simulation governance design, simulation governance navigation, simulation governance classification, simulation diagnostics, simulation architecture development, simulation governance assessment, simulation lifecycle governance, and simulation interoperability across AI systems, autonomous systems, robotics, digital twins, engineering systems, scientific models, critical infrastructure, space systems, and future simulation environments.

SIMULOS® consists of 12 Parent Standards and 60 Core Modules.

Additional standards and modules may be added when new governable simulation spaces are identified. However, the standards and modules listed below form the core structural backbone of SIMULOS®. 1. Simulation Object Standard (SOS)

Governed Space: Simulation Object

Core Question: What exactly is being simulated, and how must the Simulation Object be identified, bounded, versioned, and distinguished?

Modules:

- SOIM — Simulation Object Identification Module
 - SOBM — Simulation Object Boundary Module
 - SOVM — Simulation Object Versioning Module
 - SOCM — Simulation Object Configuration Module
 - SODM — Simulation Object Documentation Module
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2. Simulation Purpose & Scope Standard (SPSS)

Governed Space: Simulation Purpose & Scope

Core Question: Why is the simulation being performed, what must it cover, and what remains outside its simulation boundary?

Modules:

- SPM — Simulation Purpose Module
 - SSCM — Simulation Scope Module
 - SEXM — Simulation Exclusion Module
 - SACM — Simulation Applicability Context Module
 - SDPM — Simulation Depth & Precision Module
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3. Reality Representation Standard (RRS)

Governed Space: Simulation Reality Representation

Core Question: Which aspects of reality must the simulation represent, at what abstraction and fidelity, and which aspects may legitimately remain simplified or excluded?

Modules:

- RSM — Reality Selection Module
 - RAM — Reality Abstraction Module
 - RFM — Representation Fidelity Module
 - RRM — Representation Resolution Module
 - RLM — Representation Limitation Module
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4. Simulation Assumption Standard (SAS)

Governed Space: Simulation Assumptions

Core Question: Which assumptions enable the simulation, and how must their basis, dependencies, uncertainty, limitations, and consequences remain governed?

Modules:

- SAIM — Simulation Assumption Identification Module
 - SABM — Simulation Assumption Basis Module
 - SADM — Simulation Assumption Dependency Module
 - SAUM — Simulation Assumption Uncertainty Module
 - SALM — Simulation Assumption Limitation Module
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5. Simulation Model & Environment Standard (SMES)

Governed Space: Simulation Model & Environment

Core Question: How must the simulation model and simulation environment be constructed, configured, bounded, and controlled?

Modules:

- SMMM — Simulation Model Structure Module
 - SECM — Simulation Environment Configuration Module
 - SBIM — Simulation Boundary Interaction Module
 - SPHM — Simulation Parameterization Module
 - SEVM — Simulation Environment Versioning Module
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6. Simulation Input & Data Standard (SIDS)

Governed Space: Simulation Inputs & Data

Core Question: Which inputs and data may enter the simulation, and under what representativeness, transformation, temporal, and uncertainty conditions?

Modules:

- SIIM — Simulation Input Identification Module
 - SDRM — Simulation Data Representativeness Module
 - SDTM — Simulation Data Transformation Module
 - STDm — Simulation Temporal Data Module
 - SIUM — Simulation Input Uncertainty Module
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7. Simulation Scenario Standard (SSS)

Governed Space: Simulation Scenario Design

Core Question: Which scenarios must be simulated to represent expected conditions, variations, edge cases, stresses, uncertainty, and materiality

relevant alternatives?

Modules:

- SCIM — Scenario Identification Module
 - SCVM — Scenario Variation Module
 - SECM — Scenario Edge-Case Module
 - SSCM — Scenario Stress-Condition Module
 - SCCM — Scenario Coverage Module
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8. Simulation Execution Standard (SES)

Governed Space: Simulation Execution

Core Question: How must a simulation be instantiated, executed, observed, repeated, controlled, and documented?

Modules:

- SIM — Simulation Instantiation Module
 - SECM — Simulation Execution Control Module
 - SOM — Simulation Observation Module
 - SRPM — Simulation Repetition Module
 - SEDM — Simulation Execution Documentation Module
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9. Simulation Output Governance Standard (SOGS)

Governed Space: Simulation Output Governance

Core Question: What exactly did the simulation produce, how must simulation outputs be bounded and interpreted, and what may legitimately be inferred from them?

Modules:

- SOIM — Simulation Output Identification Module
 - SOCM — Simulation Output Classification Module
 - SIM — Simulation Inference Module
 - SOLM — Simulation Output Limitation Module
 - SODM — Simulation Output Documentation Module
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10. Simulation-to-Reality Correlation Standard (SRCS)

Governed Space: Simulation-to-Reality Correlation

Core Question: To what extent do simulated behavior, conditions, and outputs correspond to the reality they claim to represent?

Modules:

- SRCM — Simulation-Reality Comparison Module
 - SCCM — Simulation Correlation Criteria Module
 - SDMM — Simulation-Reality Deviation Module
 - SRAM — Reality Alignment Module
 - SCUM — Correlation Uncertainty Module
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11. Simulation Evidence & Reliance Boundary Standard (SERBS)

Governed Space: Simulation Evidence & Reliance Boundary

Core Question: What evidential meaning may simulation results legitimately carry, and where does simulation-specific reliance end before broader validation governance becomes necessary?

Modules:

- SEQM — Simulation Evidence Qualification Module
 - SEWM — Simulation Evidence Weight Module
 - SRBM — Simulation Reliance Boundary Module
 - SUM — Simulation Use Limitation Module
 - SVIM — Simulation Validation Interface Module
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12. Simulation Lifecycle & Resimulation Standard (SLRS)

Governed Space: Simulation Lifecycle, Drift & Resimulation

Core Question: How must simulation relevance be monitored over time, and when must changed reality trigger update, resimulation, replacement, or retirement?

Modules:

- SLMM — Simulation Lifecycle Monitoring Module
 - SDRM — Simulation Drift Recognition Module
 - SRGM — Resimulation Trigger Module
 - SRSM — Resimulation Scope Module
 - SRTM — Simulation Retirement & Transition Module
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SIMULOS® Architecture Summary

- Parent Standards: 12
 - Core Modules: 60
 - Core Governed Simulation Spaces
 - Simulation Object
 - Simulation Purpose & Scope
 - Simulation Reality Representation
 - Simulation Assumptions
 - Simulation Model & Environment
 - Simulation Inputs & Data
 - Simulation Scenario Design
 - Simulation Execution
 - Simulation Output Governance
 - Simulation-to-Reality Correlation
 - Simulation Evidence & Reliance Boundary
 - Simulation Lifecycle, Drift & Resimulation
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SIMULOS® Navigation Layer Core Questions Map

SOS → What exactly is being simulated, and how must the Simulation Object be identified, bounded, versioned, and distinguished?

SPSS → Why is the simulation being performed, what must it cover, and what remains outside its simulation boundary?

RRS → Which aspects of reality must the simulation represent, at what abstraction and fidelity, and which aspects may legitimately remain simplified or excluded?

SAS → Which assumptions enable the simulation, and how must their basis, dependencies, uncertainty, limitations, and consequences remain governed?

SMES → How must the simulation model and simulation environment be constructed, configured, bounded, and controlled?

SIDS → Which inputs and data may enter the simulation, and under what representativeness, transformation, temporal, and uncertainty conditions?

SSS → Which scenarios must be simulated to represent expected conditions, variations, edge cases, stresses, uncertainty, and materially relevant alternatives?

SES → How must a simulation be instantiated, executed, observed, repeated, controlled, and documented?

SOGS → What exactly did the simulation produce, how must simulation outputs be bounded and interpreted, and what may legitimately be inferred from them?

SRCS → To what extent do simulated behavior, conditions, and outputs correspond to the reality they claim to represent?

SERBS → What evidential meaning may simulation results legitimately carry, and where does simulation-specific reliance end before broader validation governance becomes necessary?

SLRS → How must simulation relevance be monitored over time, and when must changed reality trigger update, resimulation, replacement, or retirement?
Architectural Position

SIMULOS® governs the complete simulation governance lifecycle.

The architecture progresses:

Simulation Object → Simulation Purpose & Scope → Reality Representation →
Simulation Assumptions → Simulation Model & Environment → Simulation Inputs &
Data → Simulation Scenario Design → Simulation Execution → Simulation Output
Governance → Simulation-to-Reality Correlation → Simulation Evidence &
Reliance Boundary → Simulation Lifecycle & Resimulation

Together these standards govern what is being simulated, why it is being simulated, how reality is represented, which assumptions enable that representation, how the model and simulation environment are constituted, what inputs enter the simulation, which scenarios are tested, how simulation execution occurs, what outputs are produced and interpreted, how those outputs correspond to reality, what evidential meaning and reliance they may legitimately support, and when changed reality requires resimulation, replacement, transition, or retirement.

SIMULOS® therefore does not terminate when a simulation execution ends.

Its lifecycle becomes recursive when simulation-derived results enter operational use:

Simulation → Operational Use → Reality Change → Simulation Drift →
Resimulation Trigger → Resimulation Scope → Lifecycle Re-Entry → Resimulation
/ Replacement / Retirement

This recursive structure allows simulation governance to remain responsive to changes in the Simulation Object, represented reality, assumptions, models, inputs, scenarios, operational conditions, correlation basis, evidence applicability, and intended reliance. Cross-Architecture Boundary

SIMULOS® ↔ ORA™ ORA™ governs what is actually happening in Operational Reality. SIMULOS® governs how selected reality is represented and simulated.

SIMULOS® ↔ VALIDOS® SIMULOS® governs what simulation and simulation-derived evidence can legitimately represent and support. VALIDOS® governs whether the total validation basis sufficiently supports a Validation Determination and intended reliance.

SIMULOS® ↔ INTEGROS® SIMULOS® governs simulation-specific structure and lifecycle. INTEGROS® governs integrity across objects, states, evidence, and processes.

SIMULOS® ↔ ADIT® SIMULOS® governs simulation execution. ADIT® governs evidential observation, preservation, verification, and auditability of what occurred.

SIMULOS® ↔ AGA™ SIMULOS® governs simulation methodology and governed simulation conditions. AGA™ governs authority, responsibility, accountability, attribution, and decision relationships surrounding simulation activities.

SIMULOS® ↔ AIG® / ASGA™ SIMULOS® may simulate AI and autonomous systems. AIG® and ASGA™ continue to govern the behavior, authority, autonomy, constraints, and operation of the actual systems. Foundational Architecture Distinctions

Simulation Object ≠ Validation Object

Represented Reality ≠ Operational Reality

Simulation Assumption ≠ Established Fact

Model Sophistication ≠ Reality Representation Sufficiency

Input Volume ≠ Input Representativeness

Scenario Count ≠ Scenario Coverage

Simulation Execution ≠ Audit of Simulation Execution

Simulation Output ≠ Simulation Evidence

Simulation Similarity ≠ Reality Equivalence

Simulation Evidence ≠ Validation Determination

Evidence Volume ≠ Evidential Strength

High Evidential Weight ≠ Unlimited Reliance

Past Simulation Relevance ≠ Continuing Simulation Relevance

Resimulation ≠ Failure of the Original Simulation Canonical Principle

SIMULOS® does not govern Operational Reality itself, general validation sufficiency, general integrity, auditability, accountability, or the behavior and authority of actual AI and autonomous systems. SIMULOS® governs the methodological conditions under which a Simulation Object is identified, reality is represented through simulation, assumptions and simulation conditions remain explicit, simulation activity is structured and executed, outputs acquire bounded meaning, simulation-derived evidence acquires legitimate reliance boundaries, and material changes in reality trigger governed resimulation, replacement, transition, or retirement throughout the complete simulation lifecycle.

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